

## E1-02 Prior Work

source: research/prior\_work.csv — v1.4.0 (2026-09-22)

(12 fields; rendered as field dictionaries to preserve all columns)

### Row 1

|                         |                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| source_id               | PW01                                                                                                               |
| title                   | A Theory for Record Linkage                                                                                        |
| authors                 | Fellegi I.P.; Sunter A.B.                                                                                          |
| year                    | 1969                                                                                                               |
| url_doi                 | <a href="https://doi.org/10.1080/01621459.1969.10501049">https://doi.org/10.1080/01621459.1969.10501049</a>        |
| reading_scope           | bib-and-abstract-verified                                                                                          |
| problem_addressed       | Optimal decision rule for linking records across files                                                             |
| data_method             | Decision-theoretic model of comparison vectors with link/non-link/possible-link dispositions                       |
| limitations             | Assumes comparison-pair errors follow known m/u probabilities; needs calibration                                   |
| overlap_with_candidates | Theoretical basis for constrained linkage decision rule and needs-review class                                     |
| proposed_difference     | Explicit uncertainty-preserving tier (needs_review) with downstream conflict checks instead of forced dispositions |
| verifiable_location     | JASA 64(328):1183-1210                                                                                             |

### Row 2

|                         |                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------|
| source_id               | PW02                                                                                                |
| title                   | Data Matching: Concepts and Techniques for Record Linkage Entity Resolution and Duplicate Detection |
| authors                 | Christen P.                                                                                         |
| year                    | 2012                                                                                                |
| url_doi                 | <a href="https://doi.org/10.1007/978-3-642-31164-2">https://doi.org/10.1007/978-3-642-31164-2</a>   |
| reading_scope           | bibliography-verified                                                                               |
| problem_addressed       | Broad survey of modern record linkage blocking similarity and evaluation                            |
| data_method             | Survey of matching techniques blocking methods and evaluation measures                              |
| limitations             | Generic survey; no billing-domain event semantics                                                   |
| overlap_with_candidates | Background for candidate blocking and precision/recall evaluation                                   |
| proposed_difference     | Domain-specific event dictionary and lineage semantics on top of generic matching                   |
| verifiable_location     | Springer 2012                                                                                       |

### Row 3

|           |                                        |
|-----------|----------------------------------------|
| source_id | PW03                                   |
| title     | Process Mining: Data Science in Action |

|                                |                                                                                                   |
|--------------------------------|---------------------------------------------------------------------------------------------------|
| <b>authors</b>                 | van der Aalst W.                                                                                  |
| <b>year</b>                    | 2016                                                                                              |
| <b>url_doi</b>                 | <a href="https://doi.org/10.1007/978-3-662-49851-4">https://doi.org/10.1007/978-3-662-49851-4</a> |
| <b>reading_scope</b>           | bibliography-verified                                                                             |
| <b>problem_addressed</b>       | Event log analysis conformance and discovery                                                      |
| <b>data_method</b>             | Process mining over event logs with timestamps                                                    |
| <b>limitations</b>             | Assumes process logs; billing events are messier multi-source                                     |
| <b>overlap_with_candidates</b> | Framework for event normalization and reconstruction audit                                        |
| <b>proposed_difference</b>     | Explicit handling of late-arriving records and as-of truncation for point-in-time correctness     |
| <b>verifiable_location</b>     | Springer 2016 (2nd ed.)                                                                           |

#### Row 4

|                                |                                                                                                             |
|--------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>source_id</b>               | PW04                                                                                                        |
| <b>title</b>                   | A Proportional Hazards Model for the Subdistribution of a Competing Risk                                    |
| <b>authors</b>                 | Fine J.P.; Gray R.J.                                                                                        |
| <b>year</b>                    | 1999                                                                                                        |
| <b>url_doi</b>                 | <a href="https://doi.org/10.1080/01621459.1999.10474144">https://doi.org/10.1080/01621459.1999.10474144</a> |
| <b>reading_scope</b>           | bib-and-abstract-verified                                                                                   |
| <b>problem_addressed</b>       | Regression on cumulative incidence under competing risks                                                    |
| <b>data_method</b>             | Semiparametric subdistribution hazard model                                                                 |
| <b>limitations</b>             | Continuous-time hazard formulation                                                                          |
| <b>overlap_with_candidates</b> | Statistical grounding for competing payment/closure events                                                  |
| <b>proposed_difference</b>     | Discrete-time interval-based conditional probabilities suitable for irregular observation                   |
| <b>verifiable_location</b>     | JASA 94(446):496-509                                                                                        |

#### Row 5

|                                |                                                                                                             |
|--------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>source_id</b>               | PW05                                                                                                        |
| <b>title</b>                   | Nonparametric Estimation from Incomplete Observations                                                       |
| <b>authors</b>                 | Kaplan E.L.; Meier P.                                                                                       |
| <b>year</b>                    | 1958                                                                                                        |
| <b>url_doi</b>                 | <a href="https://doi.org/10.1080/01621459.1958.10501452">https://doi.org/10.1080/01621459.1958.10501452</a> |
| <b>reading_scope</b>           | bibliography-verified                                                                                       |
| <b>problem_addressed</b>       | Survival estimation with right censoring                                                                    |
| <b>data_method</b>             | Product-limit estimator                                                                                     |
| <b>limitations</b>             | Classical; no covariates                                                                                    |
| <b>overlap_with_candidates</b> | Censoring conventions adopted in v1                                                                         |
| <b>proposed_difference</b>     | H-dependent interval decomposition with explicit open-state probability                                     |
| <b>verifiable_location</b>     | JASA 53(282):457-481                                                                                        |

## Row 6

|                         |                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| source_id               | PW06                                                                                                                                |
| title                   | Deep Claim: Payer Response Prediction from Claims Data with Deep Learning                                                           |
| authors                 | Kim B.-H.; Sridharan S.; Atwal A.; Ganapathi V.                                                                                     |
| year                    | 2020                                                                                                                                |
| url_doi                 | <a href="https://arxiv.org/abs/2007.06229">https://arxiv.org/abs/2007.06229</a>                                                     |
| reading_scope           | bib-and-abstract-verified                                                                                                           |
| problem_addressed       | Denial and response-date prediction from claims                                                                                     |
| data_method             | Deep multi-task architecture over 2.9M claims from two US health systems                                                            |
| limitations             | Pre-submission features; no billing-event reconstruction; deep model not needed for this study                                      |
| overlap_with_candidates | Closest applied work on payer response prediction                                                                                   |
| proposed_difference     | Adds explicit reconstruction-quality features and uncertainty-preserving linkage as inputs; simpler transparent baseline comparison |
| verifiable_location     | arXiv:2007.06229 (ICML 2020 HSYS Workshop)                                                                                          |

## Row 7

|                         |                                                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| source_id               | PW07                                                                                                                              |
| title                   | Predicting Invoice Payment Outcomes for Optimized Collections                                                                     |
| authors                 | Zeng S.; Melville P.; Lang C.A.; et al.                                                                                           |
| year                    | 2007                                                                                                                              |
| url_doi                 | <a href="https://prem-melville.com/publications/equitant-kdd07.pdf">https://prem-melville.com/publications/equitant-kdd07.pdf</a> |
| reading_scope           | bib-and-abstract-verified                                                                                                         |
| problem_addressed       | Classify invoice payment outcome into delay buckets for collection prioritization                                                 |
| data_method             | Supervised learning over invoice-level features                                                                                   |
| limitations             | Generic B2B invoices; no claim adjudication semantics                                                                             |
| overlap_with_candidates | Baseline inspiration for delay-bucket prediction and aging features                                                               |
| proposed_difference     | Event-level observation-window aware labeling instead of static outcome buckets                                                   |
| verifiable_location     | KDD 2007 Workshop on Data Mining for Business Applications                                                                        |

## Row 8

|                   |                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| source_id         | PW08                                                                                                                                                                                  |
| title             | A machine-learning approach towards solving the invoice payment prediction problem                                                                                                    |
| authors           | Schoonbee S.                                                                                                                                                                          |
| year              | 2022                                                                                                                                                                                  |
| url_doi           | <a href="https://scielo.org.za/scielo.php?pid=S2224-78902022000400010&amp;script=sci_arttext">https://scielo.org.za/scielo.php?pid=S2224-78902022000400010&amp;script=sci_arttext</a> |
| reading_scope     | bibliography-verified                                                                                                                                                                 |
| problem_addressed | Interval-coded late-payment prediction for AR prioritization                                                                                                                          |

|                                |                                                                |
|--------------------------------|----------------------------------------------------------------|
| <b>data_method</b>             | ML classifier over invoice features into lateness intervals    |
| <b>limitations</b>             | South African industrial invoices; no censoring treatment      |
| <b>overlap_with_candidates</b> | Supports aging-stage baseline design                           |
| <b>proposed_difference</b>     | Censoring-aware competing-event labeling and refusal mechanism |
| <b>verifiable_location</b>     | South African Journal of Industrial Engineering 33(4)          |

## Row 9

|                                |                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------|
| <b>source_id</b>               | PW09                                                                                          |
| <b>title</b>                   | Waste in the US Health Care System: Estimated Costs and Potential for Savings                 |
| <b>authors</b>                 | Shrank W.H.; Rogstad T.L.; Parekh N.                                                          |
| <b>year</b>                    | 2019                                                                                          |
| <b>url_doi</b>                 | <a href="https://doi.org/10.1001/jama.2019.13978">https://doi.org/10.1001/jama.2019.13978</a> |
| <b>reading_scope</b>           | bibliography-verified                                                                         |
| <b>problem_addressed</b>       | Administrative complexity as dominant waste source                                            |
| <b>data_method</b>             | Cost estimation study                                                                         |
| <b>limitations</b>             | Not a methods paper                                                                           |
| <b>overlap_with_candidates</b> | Motivation for revenue-cycle analytics                                                        |
| <b>proposed_difference</b>     | Not applicable                                                                                |
| <b>verifiable_location</b>     | JAMA 322(15):1501-1509                                                                        |